AgentX

A simple guide to the project and its files

AgentX is a clinic front-desk assistant. Patients can ask about the clinic, find free appointments, book a slot, and request cancellations. Staff approve requests that need permission. PostgreSQL stores the records, and LangGraph manages the conversation steps.

Application and conversation files

File	Its job
app/main.py	Starts the API and connects the database, services, workflows, and endpoints.
app/config.py	Reads and checks settings such as the database address and AI model configuration.
app/llm.py	Connects to the AI model and provides a fake model for testing.
app/api/conversations.py	Provides endpoints to start conversations and send patient messages.
app/api/appointments.py	Provides endpoints to search, book, and request cancellation of appointments.
app/api/approvals.py	Provides endpoints for staff to view requests and approve or reject them.
app/api/dependencies.py	Gets the authenticated patient or staff identity for protected requests.
app/graphs/states.py	Defines the information remembered during a conversation, such as messages and booking details.
app/graphs/routing.py	Identifies the request and chooses its workflow path, checking emergencies first.
app/graphs/workflows.py	Connects the conversation steps and pauses actions that need staff approval.

Services and database files

File	Its job
app/services/conversations.py	Runs and resumes conversations while handling repeated requests safely.
app/services/appointments.py	Applies appointment rules and checks that bookings belong to the patient's conversation.
app/services/approvals.py	Saves staff decisions and resumes the related paused conversation.
app/services/clinic_knowledge.py	Creates answers from approved clinic documents and includes their sources.
app/rag/retriever.py	Searches clinic documents for passages relevant to the patient's question.
app/repositories/doctors.py	Reads doctor information from PostgreSQL.
app/repositories/appointments.py	Searches the database for available appointment slots.
app/repositories/bookings.py	Creates and reads booking records in the database.
app/repositories/conversations.py	Stores conversation records, ownership, and information for handling repeated requests.
app/repositories/approvals.py	Stores approval requests and updates bookings when staff decide.
app/repositories/models.py	Defines consistent record structures for doctors, slots, bookings, and approvals.
app/repositories/errors.py	Defines shared errors such as unavailable slots or missing records.
app/repositories/_locks.py	Controls the order of database locks during simultaneous operations.
app/db/schema.sql	Creates database tables and rules that prevent invalid or duplicate bookings.
app/db/initialize.py	Prepares the clinic database and LangGraph checkpoint tables.
app/db/seed.sql	Adds fictional example data for development.

Setup and supporting files

File	Its job
Dockerfile	Defines how to build the API's Docker container.
compose.yaml	Starts PostgreSQL, database initialization, and the API together.
.env.example	Shows which settings developers need to configure.
pyproject.toml	Defines dependencies and settings for development tools and tests.
uv.lock	Keeps dependency versions consistent across computers.
.github/workflows/ci.yml	Automatically runs code checks and tests on GitHub.
README.md	Explains the project and how to set it up and run it.
AGENTS.md	Records team responsibilities and project rules.
tests/	Contains automated checks for the API, services, database, and workflows.
docs/	Explains the architecture, research, and technical decisions.
data/clinic_documents/	Holds the clinic information used to answer patient questions.

How a request moves through the project

The API receives the request. The service applies the rules. The repository reads or updates the database. For chat requests, LangGraph chooses and runs the conversation steps.

Source and scope

Based on the GitHub code inspected on 7 November 2026, at commit 37af61c. This guide covers the main application and supporting files, with tests and documentation grouped by folder.

github.com/rahafsk/AgentX